

Geometry
Unit 12
Lesson 2 Practice

Name Answer Key
Date _____
Period _____

Circle A with secants $\overline{PM}$ and $\overline{CM}$ is shown. $\overline{PM}$ and $\overline{CM}$ intersect at point M outside the circle so that $PD = 17$, $DM = 3$, $CK = x$, and $KM = 5$.

1. What is the length of secant $\overline{PM}$?

$$PM = PD + DM = 17 + 3 = 20$$

2. Find the value of x .

$$5(x + 5) = 3(20)$$

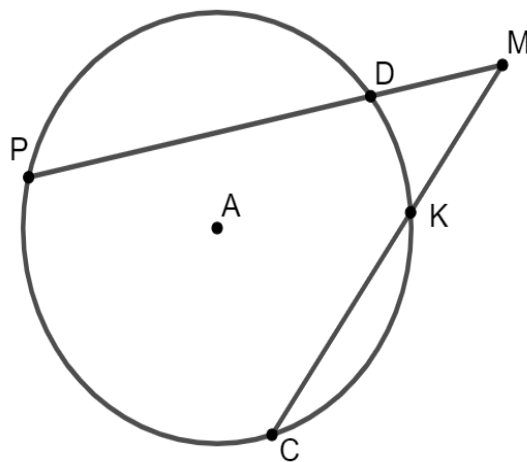
$$5x + 25 = 60$$

$$5x = 35$$

$$x = 7$$

3. What is the length of secant $\overline{CM}$?

$$CM = CK + KM = 7 + 5 = 12$$



Circle L is shown, where points G , X , A , and H lie on circle L . Secants $\overline{BX}$ and $\overline{BA}$ intersect at point B outside the circle such that $BG = 8$, $GX = (7x - 4)$, $BH = 6$, and $HA = 18$.

4. What is the value of x ?

$$8(8 + 7x - 4) = 6(6 + 18)$$

$$8(7x + 4) = 144$$

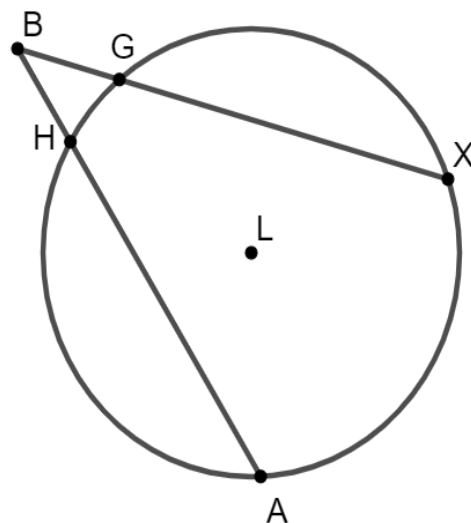
$$56x + 32 = 144$$

$$56x = 112$$

$$x = 2$$

5. Calculate the length of secant $\overline{BX}$.

$$BX = BG + GX = 8 + 7(2) - 4 = 18$$



Circle M with secants $\overline{GT}$ and $\overline{ST}$ is shown. Secants $\overline{GT}$ and $\overline{ST}$ intersect outside the circle at point T so that $GW = 7x$, $WT = 2x$, $SN = 6$, and $NT = 6$.

6. Find the value of x .

$$2x(7x + 2x) = 6(6 + 6)$$

$$14x^2 + 4x^2 = 72$$

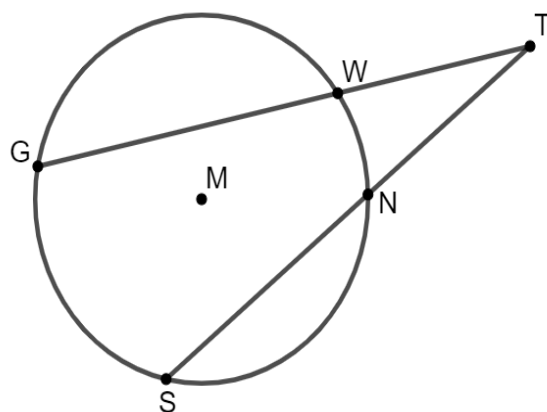
$$18x^2 = 72$$

$$x^2 = 4$$

$$x = 2$$

7. What is the length of secant $\overline{GT}$?

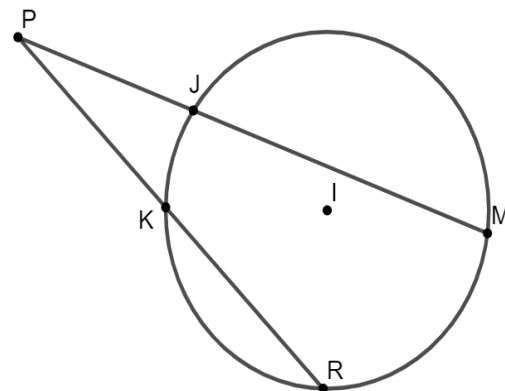
$$GT = 7x + 2x = 9(2) = 18$$



Bailey and Luciano were solving for x in circle I where points J , M , R , and K are on the circle and secants $\overline{PM}$ and $\overline{PR}$ intersect at point P outside the circle. They know $PJ = 2$, $JM = (12x - 5)$, $PK = 3$, and $KR = 4x$.

Each of their equations are set up to solve for x .

Bailey's Work	Luciano's Work
$3(4x + 3) = 2(12x - 3)$	$4x(4x + 3) = 2(12x - 3)$



8. Who wrote the correct equation needed to solve for x ?

Bailey

9. Use this equation to find the value of x .

$$3(4x + 3) = 2(12x - 3)$$

$$12x + 9 = 24x - 6$$

$$-12x = -15$$

$$x = \frac{15}{12} = \frac{5}{4} = 1.25$$

10. What is the length of secant $\overline{PR}$?

$$PR = PK + KR = 3 + (4x) = 3 + 4(1.25) = 3 + 5 = 8$$